

Week 2 Project Summary

FinTrust Digital Bank — Data Analytics Track

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1. What Was Planned

Per the Week 1 plan: investigate and document the handling of missing Device_Type/Location values, standardise Transaction_DateTime, perform EDA, calculate baseline KPI figures, and begin building the Power BI dashboard.

2. What Was Actually Completed

All five track deliverables plus two required documentation items: a Data Quality assessment (delivered as a separate PDF report and Excel cleaning log, per mentor guidance), a SQL analysis directly answering the 8 analytical questions committed to in Week 1, an 8-visualization Python EDA notebook, a Power BI dashboard (6 KPIs, 6 charts, 5 slicers, custom branded theme — final polish in progress), a 6-finding Business Insights document, and full Week 2 Documentation.

3. Tools Used

Microsoft Excel, PostgreSQL, Python (Pandas, NumPy, Matplotlib, Seaborn) in Google Colab Notebook, and Power BI.

4. Key Findings

- International transactions carry ~2x the risk-flag rate of domestic ones (36.9% vs 18.9%), and the gap holds — often widens — within every channel individually.
- Customer segment does not reliably identify high-value individual customers — several top-10 customers by spend are 'Everyday', not 'Premium'.
- ~1 in 10 transactions doesn't complete cleanly, tying up NGN49.7M in Failed/Reversed/Pending status.
- Higher transaction value predicts a higher risk-flag rate, but the effect is concentrated in the top value quartile (25.3%) rather than a smooth gradient.

5. Major Challenges Encountered

- The working environment had no PostgreSQL installed — resolved by installing and configuring it locally so every SQL query could be verified against the intern's actual working database, not just a lighter substitute engine.
- An early version of the SQL analysis answered the Week 2 brief's general categories rather than the specific 8 questions committed to in the Week 1 plan — caught during review and rebuilt to directly match those commitments.
- Power BI was a new tool with a real learning curve on relationships, DAX measures, and layout precision — addressed through a structured, stage-by-stage build process.

6. Important Decisions Made

- Missing Device_Type/Location values were filled with an explicit 'Unknown' category rather than dropped, to preserve each transaction's other valid fields for analysis.
- Statistical outliers and account-status anomalies were flagged, not deleted — both represent real signal (high-value transactions; dormant-account activity) worth investigating, not data errors to discard.
- The combined Data Quality & Cleaning Workbook was split into a PDF findings report and a separate Excel cleaning log, following explicit mentor instruction to keep the two artifacts distinct.

7. Changes to the Week 1 Approach

Scope expanded in one area: rather than only calculating baseline KPI figures, a full 8-question SQL analysis was built and verified. Scope narrowed in another: 3 of 7 planned KPI baselines (Channel Mix Share, Dormant Account Rate, Digital Engagement by Segment) were deferred to live Power BI DAX measures rather than pre-calculated separately, since they are straightforward to compute directly in the dashboard.

8. Testing / Evaluation Performed

The SQL analysis was executed end-to-end against a freshly created PostgreSQL database with primary/foreign key constraints enforced, with zero errors. The Python notebook was executed top-to-bottom with a live kernel, with zero errors. Power BI's KPI cards were cross-checked against the SQL and Python outputs and matched exactly in every case.

9. Key Limitations

- Only three months of transaction data (Jan-Mar 2026) are available, limiting any seasonality or trend analysis.
- The Digital Engagement Score finding is correlational (measured at 0.038), not a causal test.
- Root causes of Mobile App's elevated failure rate and the Dormant/Restricted-account transaction anomaly have been flagged but not yet investigated.

10. Remaining Work

Final Power BI polish: apply the custom theme, complete precise sizing/alignment, correct the monthly trend chart's line style, and capture a final dashboard screenshot for the repository. LinkedIn/X professional-development posts to follow once the dashboard is finalised.

11. Proposed Focus for Week 3

Per the project roadmap: complete and validate the Power BI dashboard, deepen the analysis, validate findings against source data, and prepare outputs for cross-track integration.